Health-a-thon 2026

FINAL SUBMISSION

Sahara

A Palliative Care Companion

TEAM NAME Zeros and Ones

TEAM CODE To confirm

Team Overview & Background

Team member	Role in prior submission	Contribution / detail
Shaik Abdul Basith	Technical lead	Product and engineering
Shaik Muhammad Awaiz	Technical member	Web and Android development
Dr Sharada Vinod Kutty	Doctor partner	Clinical workflow input
Dr Sujay Halkur Shankar	Clinician	Clinical concept and care framework
Shirin Ayub	Medical student	Team member

Team photos and individual write-ups will follow the team discussion.

Doctor specialities, experience and institutional affiliations remain to confirm.

Problem Statement and Supporting Evidence

- Families may face an emergency without a clear, current record of the patient's care preferences.
- Our clinicians identified inconsistent goals-of-care conversations, documentation and later retrieval. These are team observations, not a measured baseline.
- Sahara connects the reviewed conversation to family access and follow-up tasks.
- A clinical pilot must establish whether this improves retrieval time, understanding and continuity.

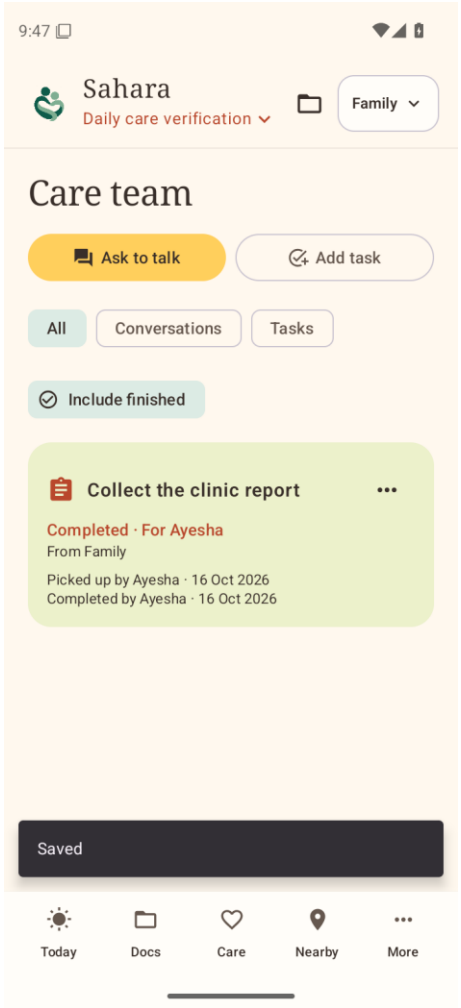
Target Users and Use Case

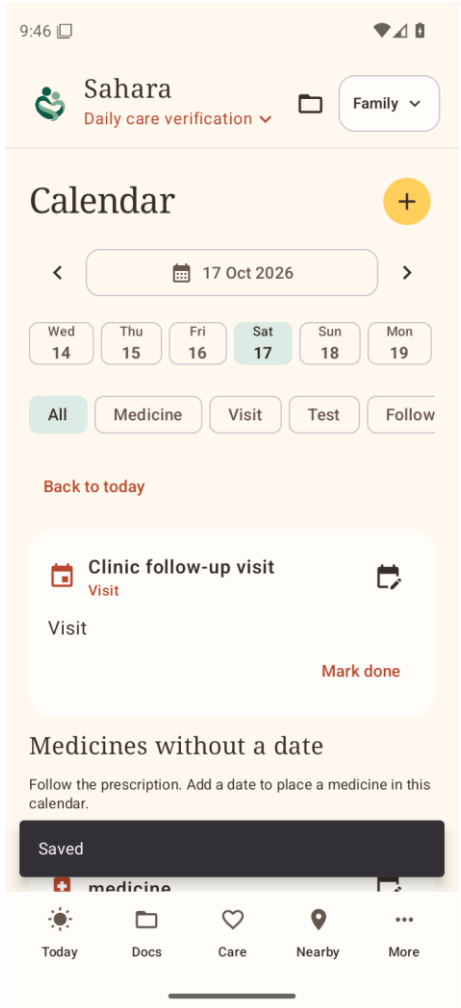
- Primary users: people with advanced cancer and their family caregivers.
- The clinician leads the goals-of-care discussion and reviews each released record.
- Clinic and home use: prepare for a conversation, retain the reviewed note and keep track of agreed care tasks.
- At a later care encounter, the patient chooses who may retrieve the current version.
- Hindi and English support aims to reduce language friction.

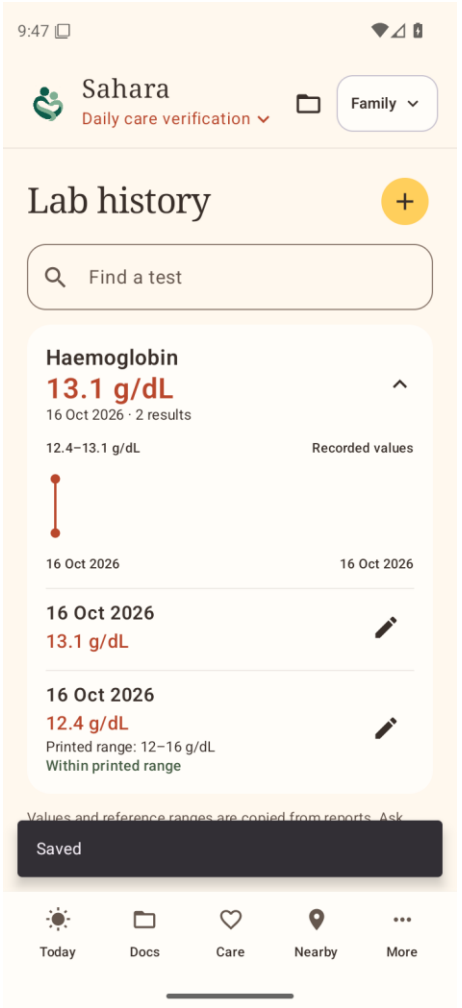
Solution Overview, Key Features & Impact - Max 2 Slides

- Record with consent or enter the conversation. Keep original words and source links.
- Review the note, release an immutable version, then capture patient and family confirmation against that exact version.
- Share by permission, with expiry and revocation. Assign follow-up tasks to named accounts.
- Daily support includes medicines, appointments, documents and a source-linked palliative-care directory.
- Expected benefit: less repeated explanation and a retrievable care record. Impact remains to be measured.

Solution Working MVP / Demo







Actual Android screens
0.5 emulator, 19 October
Fictional records

Current 0.6 test APK:
phone testing pending

Web MVP:
sahara.wedevit.in

Updated video: pending

Technology, Tools and AI Approach-

- Android: Kotlin, Jetpack Compose, Room and MapLibre Native. Web: React/Next.js with shared Supabase records.
- Optional local speech: Qwen3-ASR-0.6B int8 through sherpa-onnx. Original audio and reviewed corrections remain attributable.
- Bundled Latin/Devanagari OCR and optional on-device Hindi/English translation assist document review.
- Directory: 1,631 public listings from Pallium India and IAPC sources.
- CARE integration is proposed. Generative clinical drafting remains off.

AI output requires human review. No diagnosis or treatment recommendation.

Build Sprint Journey and Progress

- September to 3 October: clinician framework, family workflow and the original Round 1 deck.
- 9–14 October: native Android app, shared records and initial Play release work.
- 16–19 October: version confirmations, permissions, task transitions, calendar/labs and backend role checks.
- 20 October: local Hindi/English tools, national directory and a 0.6 test APK with 147 unit checks.
- Next: current phone tests, clinical language evaluation and a governed pilot.

Scope changes: stronger source attribution and local processing. Clinical drafting and CARE integration remain incomplete.

Evaluation and User Validation

- 20 October: 147 native unit checks passed. Instrumentation compiled but did not run on the current APK.
- 19 October: 284 web checks plus 65 workflow/directory database checks.
- Desktop ASR: 7.15 s English clip transcribed exactly in 5.95 s. A 9.12 s Hindi clip took 22.05 s and contained word errors.
- An 8.28 s public Hinglish clip took 22.53 s and also contained errors.
- No clinical accuracy score or patient outcome is established.

Still required: current handset tests, clinician-reviewed bilingual cases and patient/family usability validation.

Safety and Responsible AI

- Obtain recording consent and bind audio, speaker attribution and review to the correct person.
- Keep source audio/text, corrections and immutable note versions. AI may mishear words, numbers or negation.
- A clinician reviews release. Patient confirmation applies to the exact reviewed version.
- Enforce account roles and permissioned sharing with expiry/revocation. Original files stay on the phone.
- Before patient use: validate recovery/deletion, language comprehension, access controls and clinical oversight.

Deployment, Cost and Sustainability Plan

- Pilot requirements: clinical owner, agreed consent, approved users, phone checks and a supported recovery/deletion process.
- Integration: Sahara already shares approved note data across its own web and native apps. CARE/EHR integration remains proposed.
- One-time budget: device checks, onboarding, clinician validation and integration. Costing remains to confirm.
- Ongoing budget: hosting, database, storage, support and governance. Local inference has no per-call cloud AI fee.
- Sustainability proposal: facility/NGO/government support. Patient access and final pricing require team agreement.

What Next? Plans to take initiative forward (if any)

- Complete Android phone checks using fictional inputs and clinician-reviewed Hindi/English cases.
- Finish signup/recovery and independent clinician verification before patient use.
- Agree a pilot setting, baseline measures and consent/oversight with clinical partners.
- Prototype a CARE clinician extension and test patient and consent workflows.
- Public sharing of solution information on Health-a-thon or other websites: Yes.

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Thank you

Zeros and Ones • Team code to confirm

workwithdevit@gmail.com
MVP: https://sahara.wedevit.in
Demo video: updated link pending
Code: github.com/basithladdu/healthathon (private)

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